

YASHAS CHANDRA BATHINI AI Engineer | Full-Stack Developer | LLM & RAG Systems | Azure & CI/CD

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 Portfolio: yashas144.github.io/web  F-1 OPT Eligible From June 2026

PROFILE

AI Engineer with 4 years of experience building scalable LLM-powered applications and RAG systems, backed by strong full-stack expertise in React and FastAPI. Delivered end-to-end AI platforms including intelligent predictive maintenance systems, recommendation engines and real-time analytics systems. Deep expertise in hybrid retrieval (BM25 + vector search), NLP pipelines, and explainable AI. Proven ability to design high performance APIs, deploy cloud-native systems, and ship responsive, user-centric products.

SKILLS

- Generative AI & LLMs:** Transformers, Hugging Face, RAG, Agentic AI, LangChain, NLP, FAISS, Chroma, Pinecone.
- Full-Stack Web Development:** React, FastAPI, REST APIs, LLM Integration, scalable API development.
- Programming:** Python, Java, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn.
- Data Engineering:** Data Analysis, Data Visualization, Tableau, MySQL, PostgreSQL, Spark, Databricks, Mosaic AI
- MLOps:** MLflow, Time-Series Forecasting (LSTM), Anomaly Detection (Isolation Forest), XGBoost, Regression, Clustering.
- Cloud & DevOps:** Azure, AWS, CI/CD Pipelines, GitHub Actions, Docker, Git, Jira.

WORK EXPERIENCE

AI Engineer 06/2022 – 05/2026
Sysarket Hyderabad, India

- Key Contributions**
- Designed and delivered AI-powered applications spanning predictive maintenance, recommendation systems, generative AI research, and sports analytics.
 - Built scalable full-stack applications using React, FastAPI, Python, Azure, and modern AI frameworks.
 - Developed production-grade APIs, retrieval systems, machine learning pipelines, and cloud-native deployments supporting real-world business use cases.
 - Collaborated across the full software lifecycle including architecture, development, deployment, monitoring, and optimization.

- Auravance - The Future of Industrial Foresight**
- Built a full-stack AI platform (React, FastAPI) for real-time equipment monitoring and failure prediction using XGBoost and LSTM, improving detection accuracy by 30%.
 - Developed a RAG-based root cause analysis system over equipment manuals, generating explainable diagnostics.
 - Designed RUL (Remaining Useful Life) prediction and anomaly detection (Isolation Forest) to reduce downtime risk.
 - Engineered scalable real-time IoT data pipelines using MQTT and Python to ingest, process, and store high-frequency sensor telemetry, enabling anomaly detection, predictive forecasting, and operational insights.

- AI-Powered Music Intelligence Platform**
- Designed and delivered an AI-powered music recommendation platform leveraging Hybrid-RAG (BM25 + vector search) across 500K+ songs, enabling context-aware discovery based on mood, genre, and user intent.
 - Architected a scalable RAG pipeline with embedding-based retrieval and ranking, significantly improving recommendation relevance and reducing noise in outputs.
 - Engineered real-time features, including real-time mood detection and seamless playback.
 - Deployed a cloud-native full-stack system (React + FastAPI) on Azure and Hugging Face with CI/CD pipelines, ensuring reliable and scalable delivery.

- Deep Learning-Based Image Synthesis with GAN Architecture | Research Publication**
- Developed GAN-based image synthesis systems using PyTorch and TensorFlow, improving output quality through architecture tuning and training optimization.
 - Contributed to applied generative AI research, developing GAN architectures for real-world applications and publication-ready experimentation.

- FIFA Analytics & Prediction System**
- Developed a robust sports analytics platform featuring ML-based match predictions, APIs, and dashboards.
 - Improved model accuracy by 18% using feature engineering, hyperparameter tuning, and MLflow tracking.
 - Implemented model registry, versioning, and monitoring (performance, confidence, and feature drift).
 - Deployed a containerized full-stack application using React (frontend) and Flask (backend) with integrated ML services, leveraging Docker and Azure with Azure SQL DB for scalable, production-ready model serving.

EDUCATION

Master of Science - Data Science 08/2024 – 05/2026
University of North Texas, GPA: 4.0 / 4.0 Denton, Texas

Bachelor of Technology - Information Technology 08/2020 – 05/2024
Jawaharlal Nehru Technological University, Hyderabad, CGPA: 3.7 / 4 Hyderabad, India

CERTIFICATIONS

Generative AI Fundamentals - Databricks | Machine Learning with Big Data - UC San Diego | Introduction to Data Science - Cisco | Career Essentials in Generative AI - Microsoft | Python Essentials - Cisco